CoverMap

Which acres of your field are generating the erosion? CoverMap maps cover crop protection at the pixel level — so the conversation moves from "did you plant a cover crop" to "did it work where it needed to."

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Sentinel-2 · 10 m | Google Earth Engine | RUSLE C × LS | Iowa NRCS FOTG

OVERVIEW

What CoverMap Does

Answers the question every cover crop producer eventually asks: *what did this actually do for my ground?*

- Maps erosion Risk Index zones at pixel resolution using satellite NDVI and Iowa 3m slope data
- Estimates C-factor reduction and soil saved (t/ac/yr) versus a no-cover baseline — by landscape position
- Generates a field report in under two minutes — Producer summary, CCA advisory report, or GIS shapefile export
- Works on any Iowa field with a boundary — no proprietary data, no site visit required

The Problem It Solves

- Producers can measure seed cost to the dollar. They cannot measure erosion reduction to the acre
- A RUSLE2 run gives one number for a field — CoverMap tells you which 40 acres of that 200-acre field are generating 80% of the loss
- Cover crop value is argued, not demonstrated — CoverMap produces the demonstration

Who It Is Built For

- **Producers** — erosion reduction value of their cover crop investment, field by field
- **Agronomists and CCAs** — spatial data and a signed report behind every conservation recommendation
- **Seed dealers and aerial applicators** — documented practice outcomes for every customer
- **Conservation organizations, extension, and researchers** — field-scale evidence linking cover crops to measurable erosion reduction across Iowa

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Workflow at a Glance

Sentinel-2 L2A
10 m surface reflectance spring window composite

→

NDVI
Per-pixel NDVI cloud-masked

→

C-Factor
Piecewise exponential by residue system

→

LS-Factor
Iowa 3 m DEM McCool et al. 1987

→

Risk Index
C × LS per pixel 4-zone classification

→

Field Report
CCA advisory + RUSLE soil loss

REMOTE SENSING

NDVI — Quantifying Cover Density from Space

The Normalized Difference Vegetation Index exploits the distinct spectral signatures of photosynthetically active tissue to estimate the density of green plant cover at 10 m resolution across an entire field in a single image acquisition.

The Physics

Chlorophyll a and b strongly absorb incident radiation in the red band (650–680 nm) for photosynthesis, while the spongy mesophyll cell structure of healthy leaves scatters and reflects strongly in the near-infrared (NIR, 800–900 nm). Bare soil and crop residue lack this differential — they reflect both bands at similar, moderate levels. NDVI amplifies this contrast:

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NDVI = (NIR − Red) / (NIR + Red) Scale: −1 to +1 Bare soil / residue: 0.05 – 0.18 Sparse cover: 0.18 – 0.30 Adequate stand: 0.30 – 0.50 Dense canopy: 0.50 – 0.80+
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Sentinel-2 Band 8 (842 nm NIR) and Band 4 (665 nm Red) are used. Both are native 10 m resolution — no pan-sharpening or resampling required. The Level-2A product delivers atmospherically corrected surface reflectance, removing most aerosol and water vapor effects before NDVI computation.

Imagery Acquisition

CoverMap queries Google Earth Engine for all Sentinel-2 L2A scenes intersecting the field boundary within a user-defined spring window (default March 15 – May 15). The Sentinel-2 constellation revisits Iowa approximately every 5 days under clear conditions.

Cloud masking: The ESA Scene Classification Layer (SCL) is applied at the pixel level to remove cloud, cloud shadow, and saturated pixels before compositing. Scenes with fewer than 50% valid pixels trigger a reliability warning; fewer than 75% valid pixels trigger a quality advisory.

Temporal composite: Valid pixels across all scenes in the window are summarized as a temporal median — reducing noise from single-date anomalies and partial cloud cover. The composite represents integrated canopy conditions across the observation window, not a single snapshot.

> **Known limitation:** NDVI saturates above approximately 3,500 kg/ha dry biomass (~3,100 lb/acre — the national cereal rye database mean). CoverMap cannot differentiate between a good and an excellent stand above this threshold. Additionally, an NDVI reading of 0.25 in late March (dormant rye) reflects different biomass than the same reading in early May (active growth). No growing degree day adjustment is currently applied.

NDVI Cover Quality Zone Classification

NDVI Range	Zone	Biomass Estimate (cereal rye)	NRCS Practice 340 Context
< 0.20	Low Cover	< 900 lb/acre	Below minimum stand threshold; inadequate erosion protection
0.20 – 0.35	Marginal	900 – 2,200 lb/acre	Borderline; may meet minimum on favorable soils; field verification recommended
> 0.35	Good Cover	> 2,200 lb/acre	Exceeds NRCS 340 stand requirement; effective erosion control likely

Biomass estimates derived from Huddell et al. (2024) national cereal rye NDVI-biomass database (n = 5,695 field observations, mean 3,428 kg/ha). Uncertainty ±40% at the field scale.

MODEL FOUNDATION

The Residue Baseline — What NDVI Actually Sees in Spring

A critical step in translating satellite NDVI to erosion-relevant C-factor is distinguishing the living cover crop signal from the background NIR reflectance of crop residue — a distinction the raw NDVI value alone cannot make.

The Problem with Raw NDVI

In March and April, an Iowa field's NDVI reading is not a pure function of living cover crop biomass. Corn stover at 80% surface coverage reflects NIR at levels that drive NDVI readings of 0.12–0.18 independently of any cover crop presence. A conventionally tilled field with no cover crop planted but residue present may produce NDVI near 0.10–0.14. Treating that reflectance as evidence of "some cover" and applying a corresponding C-factor reduction would overstate erosion protection.

Any C-factor model that begins awarding living-cover credit at NDVI = 0 implicitly assumes that the entire NDVI signal originates from living vegetation — an assumption that fails in Iowa spring conditions across every tillage system.

The Universal NDVI Baseline: 0.185

CoverMap establishes a residue signal threshold below which no living-cover credit is awarded. Below NDVI = 0.185, C-factor equals the residue system's intercept — the erosion rate for that tillage system's residue alone, with no established cover crop. Above the threshold, C-factor decays exponentially as living plant biomass adds measurable canopy protection on top of the residue layer.

Baseline Derivation

Parameter	Value
Observations	15 Iowa fields
Counties	5 Iowa counties
Tillage systems	No-till, conservation, conventional
Previous crops	Corn and soybean
Imagery	Sentinel-2 L2A, March–April 2026
Mean bare/residue NDVI	0.179
Standard deviation	0.013
Threshold (mean + 0.5 SD)	**0.185**

Threshold derived from n=15 Iowa no-cover-crop field observations across five counties (internal calibration dataset, spring 2026; mean NDVI = 0.179, SD = 0.013). No statistically significant difference in residue NDVI baseline was found across residue type or tillage system — supporting a single universal threshold.

Agronomic interpretation: An Iowa field with mean NDVI of 0.18 in April is not demonstrating cover crop stand — it is demonstrating residue. The cover crop may be present but too sparse or dormant to register above the soil + residue background. CoverMap's 0.185 threshold prevents that residue signal from being credited as erosion protection the cover crop has not yet earned.

RUSLE COMPONENT 1

C-Factor — Piecewise Exponential Model by Residue System

The cover-management factor C encapsulates all vegetation and tillage effects on erosion relative to a standard bare fallow (C = 1.0). CoverMap parameterizes C as a piecewise exponential function of NDVI, differentiated by the CCA-selected residue system.

Model Formula

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Residue-only zone (no living-cover credit): C(NDVI) = intercept where NDVI ≤ 0.185 Living-cover zone (exponential decay): C(NDVI) = floor + (intercept − floor) × exp(−k × (NDVI − 0.185)) where NDVI > 0.185 Parameters by residue system: intercept = C at NDVI = 0 (residue protection only, no cover crop) floor = C asymptote at high NDVI (maximum cover crop benefit) k = decay constant (higher = faster benefit per unit NDVI)
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Parameter Table

Residue System	intercept	floor	k
No-till corn (~80% residue)	0.05	0.005	8
No-till soybeans (fragile residue)	0.10	0.015	7
Conservation tillage (>30% residue)	0.25	0.050	6
Conventional tillage (<30% residue)	0.45	0.100	5
Unknown (conservative default)	0.45	0.100	5

Parameters are initial estimates based on published RUSLE2 value ranges.

Reading the Parameters

Intercept is the C-factor for a field with that tillage system and zero living cover crop. No-till corn at intercept = 0.05 reflects that 80% corn stover coverage achieves substantial erosion protection independent of cover crop establishment — a well-documented field observation. Conventional tillage at 0.45 reflects near bare-soil conditions with minimal residue, consistent with NRCS RUSLE2 guidance.

Floor is the best achievable C under near-canopy-saturation NDVI for that system. No-till corn reaches a floor of 0.005 — effectively trace erosion relative to bare soil. Conventional tillage floor = 0.10 reflects irreducible risk from tillage-induced surface roughness and reduced aggregate stability even under dense living cover.

k governs response speed. No-till corn (k = 8) decays fastest — even modest green cover on top of a strong residue base yields large proportional C reductions. Conventional tillage (k = 5) responds more slowly because each unit of NDVI gain must overcome a larger initial C.

C-Factor Curves by Residue System

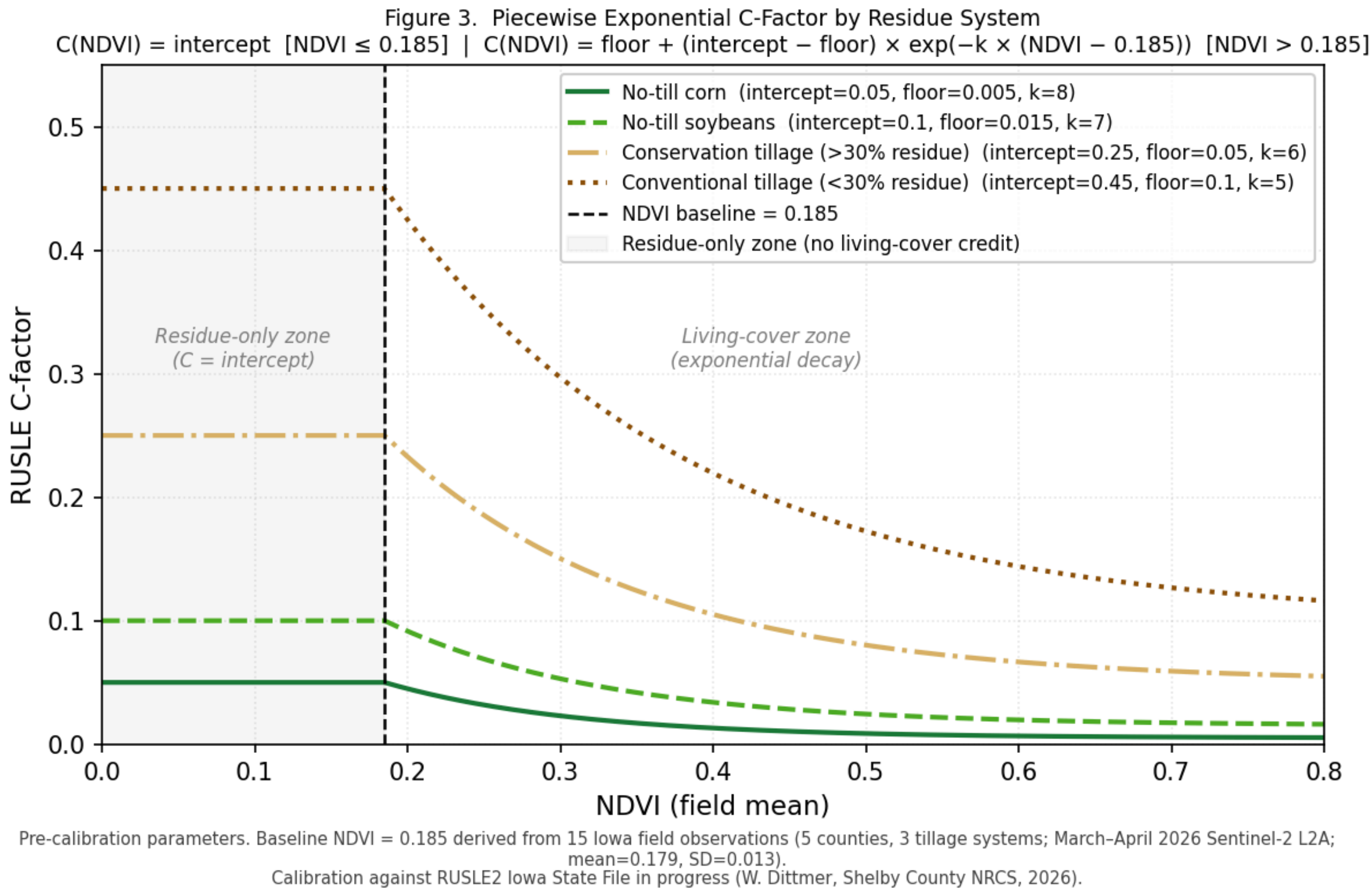

Figure 3. Piecewise exponential C-factor curves. The flat region left of the dashed line (NDVI ≤ 0.185) awards no living-cover credit — C equals the residue-system intercept. Above 0.185, C decays exponentially toward the floor asymptote. Pre-calibration parameters.

Why the curves don't start at zero: Even with no cover crop, crop residue and tillage-induced roughness reduce erosion relative to a bare tilled reference. The intercept encodes that protection. The cover crop's job is to push C from the intercept down toward the floor — the additional reduction visible in the decaying portion of each curve.

RUSLE COMPONENT 2

LS-Factor — Terrain Amplification of Erosion Risk

The LS-factor is the topographic multiplier on erosion. It is why a cover crop at NDVI 0.25 on a 3% sideslope is not the same erosion story as the same cover crop on the 14% backslope 200 feet upslope.

The McCool Formula

CoverMap computes S-factor using the continuous McCool et al. (1987) piecewise formula, calibrated to natural slope gradients:

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θ = arctan(slope% / 100) S-factor (steepness): S = 10.8 × sin(θ) + 0.03 slope < 9% S = 16.8 × sin(θ) − 0.50 slope ≥ 9% S ≥ 0.03 floor prevents zero on flat ground L-factor (slope length, λ = 100 m assumed): L = (100 / 22.13) ^ m m = 0.2 (slope < 1%) m = 0.3 (1% ≤ slope < 3%) m = 0.4 (3% ≤ slope < 5%) m = 0.5 (slope ≥ 5%) LS = L × S (computed at every 10 m pixel)
```

> **Slope length assumption:** True RUSLE2 LS requires horizontal distance from flow origin to concentration point — a flow-accumulation computation. CoverMap uses a fixed 100 m slope length for all pixels, consistent with typical Iowa row-crop slope segment geometry. This introduces conservatism on long uniform slopes (>200 m) and may underestimate LS on short steep concentrated-flow segments. Appropriate for field advisory use; not a substitute for formal NRCS RUSLE2 analysis.

LS Values at 100 m Slope Length

Slope (%)	LS-Factor	Risk Amplification vs. 4%
1	0.14	0.14×
2	0.28	0.28×
4	0.73	1.0× (reference)
6	1.21	1.7×
9	2.06	2.8×

Slope (%)	LS-Factor	Risk Amplification vs. 4%
12	2.91	4.0×
15	3.75	5.1×
18	4.59	6.3×

> **Why per-pixel LS matters:** A field with mean slope of 8% and a 16% backslope generates LS-factors of 1.8 and 4.3 respectively. Applying the mean slope to the whole field understates backslope risk by more than 2×. CoverMap computes LS independently at every 10 m pixel using the Iowa DNR 3 m LiDAR-derived DEM, then resamples to match the Sentinel-2 NDVI grid before multiplying with C-factor.

CLASSIFICATION

Risk Index and Zone Classification

The per-pixel Risk Index is the product of C-factor and LS-factor — a dimensionless erosion potential that CoverMap classifies into four advisory zones. Every 10 m pixel in the field boundary receives an independent zone assignment based on its own NDVI and slope values.

```
Risk Index = C(NDVI, residue system) × LS(slope%) Computed at every 10 m pixel within the field boundary mask. Higher values = higher erosion potential from the combination of poor cover and steep terrain.
```

Low Risk

`Risk Index < 0.3`

Cover crop providing effective protection. Erosion rate well below tolerable loss for most Iowa soil types under normal spring rainfall.

Moderate Risk

`0.3 – 0.7`

Variable protection. Stand may be adequate on gentler ground but marginal on steeper slope positions. Field verification recommended prior to termination.

High Risk

`0.7 – 1.5`

Insufficient cover for slope conditions. These pixels are generating measurable soil loss above tolerable rates. Cover crop termination should be delayed if possible.

Critical Risk

`Risk Index > 1.5`

Severe erosion potential. Combination of poor cover and steep slope generates risk comparable to bare conventional tillage on moderate terrain. Immediate agronomic attention warranted.

> **Field-average vs. per-pixel — why the distinction matters:** A 48-acre Shelby County field with mean NDVI 0.25 and mean slope 8% might appear Moderate overall. But within that same field, the 16-acre backslope unit at NDVI 0.15 and slope 14% generates Critical-level Risk Index values — and that backslope is where Iowa loses the most topsoil per unit area per rain event. CoverMap's per-pixel approach surfaces that spatial heterogeneity rather than averaging it away.

EXAMPLE OUTPUT

Reading a CoverMap Field Report — Bin Field, Shelby County

The following three-page report was generated for a conservation tillage field in Shelby County, Iowa using April 2026 Sentinel-2 imagery. Walk through each section to understand what CoverMap measures and how to interpret the advisory output.

Page 1 — Field Maps and Metadata

CoverMap

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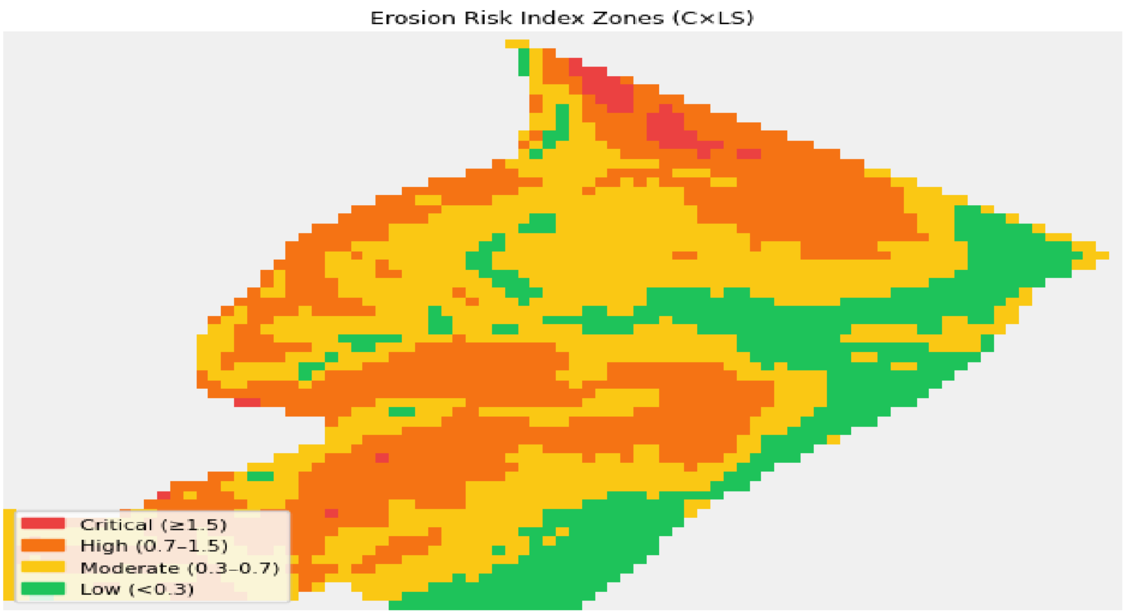
Field Cover Crop Assessment

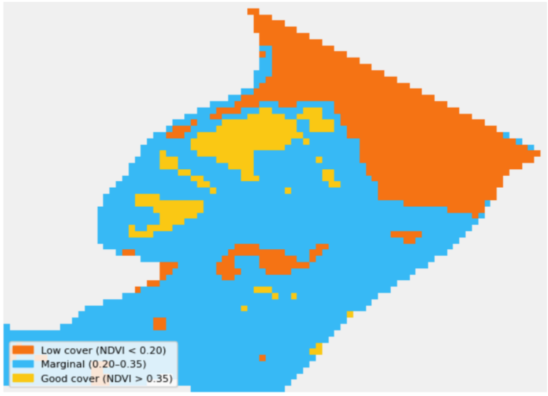
Field: Bin Field (2)	**Farm:** Bizy Farms Inc.	**County:** Shelby County, IA	**Report Date:** May 15, 2026
Previous crop: Corn	**Termination date:** May 1st.	**Dominant soil series:** Monona — K-factor: .32	**Previous crop / tillage:** Tillage — > 30% residue (conservation tillage)

NDVI: Apr 05, 2026 – Apr 20, 2026 | DEM: Iowa 3-meter Digital Elevation Model (Iowa DNR)

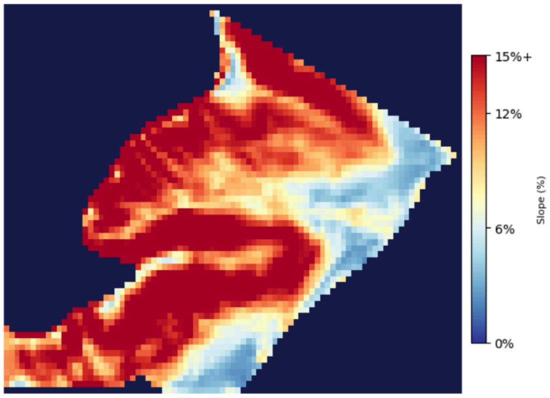
Field Risk Maps



Erosion Risk Index Zones (C×LS) — pixel-level RUSLE risk classification



NDVI Cover Quality — Low (<0.20) / Marginal / Good (>0.35)



Terrain Slope (% gradient) — Flat / Moderate / Steep

Page 1: Field identification, satellite acquisition metadata, and three spatially co-registered maps — Erosion Risk Index, NDVI Cover Quality, and Terrain Slope. All maps share the same 10 m pixel grid and field boundary mask.

Erosion Risk Index Map

The primary output. Each pixel is colored by its C × LS Risk Index value — the combination of cover density (from NDVI) and slope steepness (from the Iowa 3 m DEM). Red and orange pixels identify where cover crop stand is insufficient relative to the terrain's erosion amplification. This is the map that answers: "*Where is this field most vulnerable right now?*"

NDVI and Slope Maps

The two component maps shown alongside the Risk Index allow the user to decompose the risk driver at any location. A High Risk pixel that is red on the NDVI map (low cover) but not particularly steep indicates a cover crop stand failure problem. A High Risk pixel that is green on the NDVI map but red on the slope map indicates a terrain-driven problem that better cover crop management can only partially offset.

Page 2 — Advisory, Zone Tables, and Field Metrics

CoverMap Advisory & Recommendation

Erosion Concern: Moderate

Cover crop stand is variable across slope positions based on NDVI analysis. Erosion risk increases on steeper units during spring rainfall events.

Cover Crop Stand — NDVI Zone Summary

Zone	Acres	% of Field
Low Cover (NDVI < 0.20)	12.9	27%
Marginal (NDVI 0.20–0.35)	31.9	66%
Good Cover (NDVI > 0.35)	3.6	7%
Total	**48.3**	**100%**

Erosion Risk Zone Summary (C×LS)

Zone	C×LS Range	Acres	% of Field
Critical Risk	> 1.5	0.7	2%
High Risk	0.7–1.5	17.2	36%
Moderate Risk	0.3–0.7	21.0	43%
Low Risk	< 0.3	9.4	19%

Risk Index = C-factor (NDVI) × LS-factor (slope). Boundary-masked field pixels only. Critical >1.5 · High 0.7–1.5 · Moderate 0.3–0.7 · Low <0.3

NDVI imagery dated 6 scenes composited: Apr 08, 2026 – Apr 18, 2026. Field conditions may have changed since image capture. This report documents satellite-observed conditions only.

Cover Crop Stand Assessment — Satellite Documentation

Requirement	Data Source	Status
Cover crop present	Sentinel-2 NDVI > 0.20	■ NDVI 0.251 — cover crop confirmed
Image date	GEE metadata	6 scenes composited: Apr 08, 2026 – Apr 18, 2026
Estimated biomass	NDVI proxy	~700–1650 lb/acre (±40% NDVI proxy)
30% ground cover	NDVI threshold	■ Estimated adequate cover zones based on NDVI threshold — field verification recommended

Satellite-verified cover crop status. Field verification recommended before termination.

Field Level Results

Metric	Value
NDVI Mean	0.251
NDVI Range	0.146 – 0.576
Slope Mean (%)	12.0%
C-Factor (exp. model)	0.185 (26% reduction vs. baseline)
Risk Index (C×LS)	0.592

Page 2: Erosion concern level with rationale, NDVI zone summary, Risk Index zone summary, and stand documentation checklist for EQIP Practice 340 compliance review.

0.251
Mean field NDVI
Apr 5–20, 2026

12.0%
Mean field slope
Iowa DNR 3m DEM

0.185
C-Factor
26% reduction vs. baseline

0.592
Risk Index (C×LS)
Moderate concern

27%
Field area in
Low Cover zone

23.4%
Est. erosion reduction
vs. no-cover baseline

NDVI Zone Breakdown

Zone	Acres	% of Field
Low Cover (NDVI < 0.20)	12.9	27%
Marginal (0.20–0.35)	31.9	66%
Good Cover (> 0.35)	3.6	7%
Total	**48.3**	**100%**

66% of the field is in the Marginal NDVI zone — cover crop present but below the Good Cover threshold. This is the spatial story CoverMap tells: the same NDVI distribution produces a Moderate Erosion Concern under conservation tillage because the residue system's C-factor intercept (0.250) is five times higher than no-till corn (0.050). The cover crop is helping — but on 12% Monona backslopes, conservation tillage needs it to do more.

Risk Index Zone Breakdown

Zone	Acres	% of Field
Critical (> 1.5)	0.7	2%
High (0.7–1.5)	17.2	36%
Moderate (0.3–0.7)	21.0	43%

Zone	Acres	% of Field
Low (< 0.3)	9.4	19%

38% of field area — 17.9 acres — classifies as High or Critical Risk. These acres are concentrated on the steeper Monona backslope positions where slope exceeds 10% and NDVI falls in the Low or Marginal zone. Despite a living cover crop present across most of the field, conservation tillage's higher C-factor intercept (0.250) means the Risk Index remains elevated on terrain-driven pixels.

Page 3 — Erosion Reduction Estimates

Erosion Concern	**Moderate**

Metric	Value
Est. Cover Crop Erosion Reduction	23.4%
Baseline Soil Loss (no cover)	46.0 t/ac/yr
Est. Soil Saved (area-weighted)	10.8 t/ac/yr

Estimates based on RUSLE C-factor methodology. C-factor derived from piecewise exponential NDVI model. ±10 pt uncertainty on reduction percentage.

NDVI Source: Sentinel-2 via Google Earth Engine (NDVI: Apr 05, 2026 – Apr 20, 2026) | DEM: Iowa 3-meter Digital Elevation Model (Iowa DNR) | Slope: computed in UTM meters (EPSG:26915)

C-Factor methodology: piecewise exponential NDVI model · continuous C-factor differentiated by residue system. This report is advisory only and does not constitute an official NRCS determination.

CoverMap Field Report · Stephen Zimmerman, CCA MS · Sentinel-2 via Google Earth Engine · Iowa RUSLE C-factor calibration · May 15, 2026

Page 3: RUSLE-based soil loss estimates, per-zone erosion reduction breakdown, comparison to soil loss tolerance (T-value), and data source citations.

RUSLE ESTIMATION

Soil Loss Estimation — Bin Field Walkthrough

When SSURGO K-factor data are available, CoverMap estimates absolute annual soil loss using simplified RUSLE. The calculation uses field-mean values for a representative estimate — it is not a substitute for formal NRCS RUSLE2 analysis on the critical slope area.

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A = R × K × LS × C × P R = 175.0 MJ·mm/ha·hr·yr Shelby County · Iowa NRCS FOTG Figure 2 (September 2002) K = 0.32 t·ha·hr/ha·MJ·mm Monona silt loam · SSURGO (SDM Data Access API) LS = 3.29 (dimensionless) mean field LS · Iowa DNR 3m DEM · McCool et al. 1987 C = 0.185 (dimensionless) cover crop C · piecewise exponential model (this report) P = 1.0 no support practice factor applied
A_current = 175 × 0.32 × 3.29 × 0.185 × 1.0 ≈ 34.1 t/ac/yr A_baseline = 175 × 0.32 × 3.29 × 0.250 × 1.0 ≈ 46.1 t/ac/yr C_baseline = intercept for conservation tillage (>30% residue) = 0.250 Soil saved = A_baseline − A_current = 10.8 t/ac/yr (23.4% reduction) T-value (Monona) = 5 t/ac/yr — field is 9.2× above T without cover · 6.8× with cover
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Interpreting the Numbers

The 46.1 t/ac/yr baseline reflects what this conservation tillage field would lose without a living cover crop — Shelby County R = 175, Monona K = 0.32, 12% mean slope, and a conservation tillage intercept of 0.250. That baseline is 9.2× the Monona T-value of 5 t/ac/yr. The cover crop reduces that to an estimated 34.1 t/ac/yr — still 6.8× T. On Monona backslopes under conservation tillage, cover crops are a meaningful tool but not a complete solution.

This is the agronomically honest output CoverMap produces: the cover crop is working — 23.4% reduction, 10.8 t/ac/yr saved — but the Risk Index map tells the CCA exactly which 38% of the field is still generating High or Critical erosion risk despite it. That spatial answer is what drives a productive conversation about tillage system, drainage, or structural practices on the critical slope positions.

How CoverMap Compares to RUSLE2

Parameter	RUSLE2	CoverMap
Slope area	Critical slope segment	Field mean

Parameter	RUSLE2	CoverMap
Slope length	Flow accumulation	100 m assumed
C-factor	Daily through crop stages	Single NDVI composite
LS precision	High (GIS-derived)	Moderate (fixed L)
Typical result	Higher (critical area)	Lower (field average)

CoverMap estimates will almost always be lower than RUSLE2 for the same field. RUSLE2 focuses on the dominant critical slope area; CoverMap averages across the entire field including flat toeslopes. Both are correct for what they measure. CoverMap reports include this disclaimer explicitly.

LITERATURE

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Sentinel-2 imagery via Google Earth Engine · Iowa RUSLE C-factor methodology · Iowa DNR 3m DEM